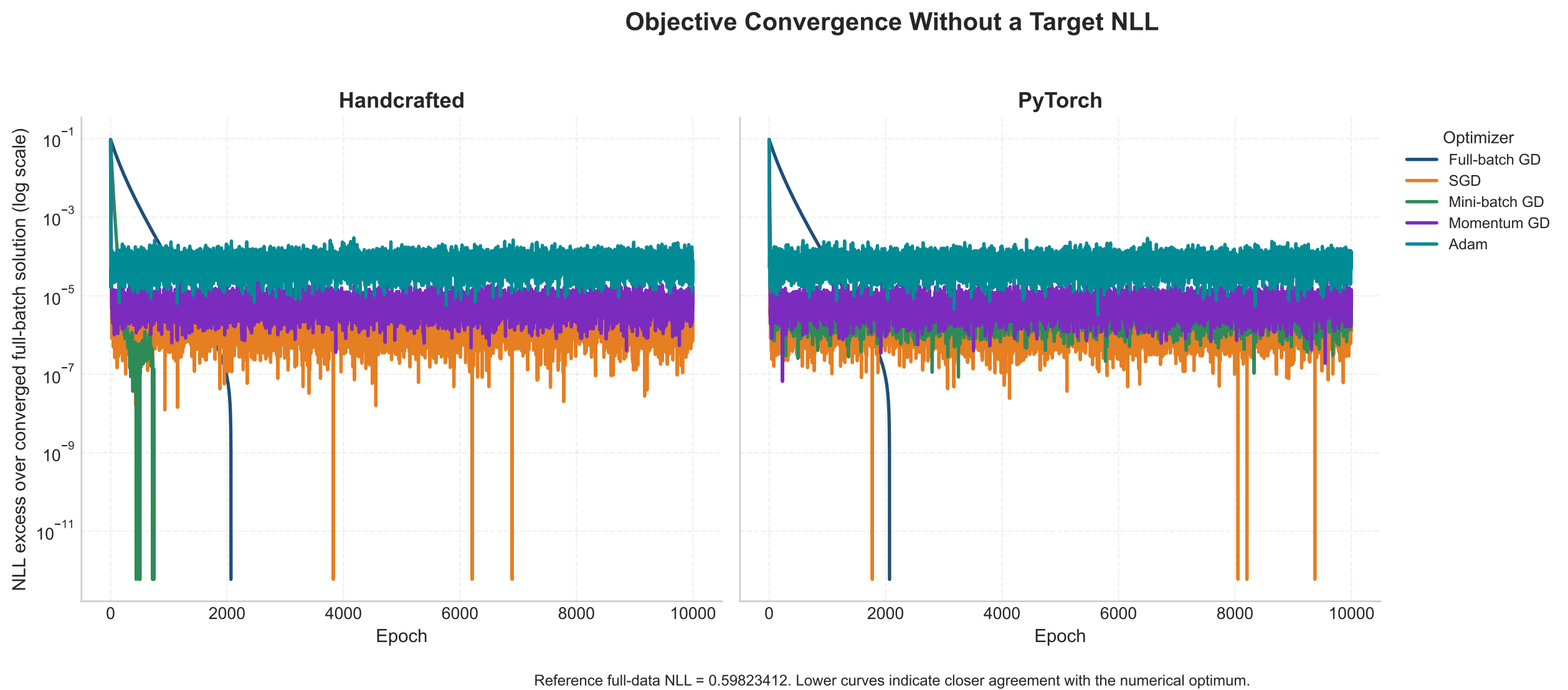

Objective Convergence Without a Target NLL
Handcrafted
PyTorch
NLL excess over converged full-batch solution (log scale)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
0
2000
4000
6000
8000
10000
Epoch
Optimizer
Full-batch GD
SGD
Mini-batch GD
Momentum GD
Adam
Reference full-data NLL = 0.59823412. Lower curves indicate closer agreement with the numerical optimum.